

Q9 Ans: B

Explanation

for orbiting satellite:

gravitational force provides centripetal force

$$\frac{GMm}{r^2} = mr \left(\frac{2\pi}{T} \right)^2$$

$$T^2 = \frac{4\pi^2}{GM} r^3$$

the radius r of orbit is independent of the mass m of the satellite

so the radius is the same

Q10 Ans: B